

LEGO® SPIKE™ Prime — Project Ideas

A pick-your-build list for Ryan — official robot designs, advanced builds, and themed models. Type or scan a link for the full instructions.

25 projects · Live gallery: https://borenw.github.io/spike_prime_projects/

1. Robot designs & build instructions (great starting points)

Free step-by-step PDFs from primelessons.org.

1. **Coop Bot** — Chicken-style walker
<https://primelessons.org/RobotDesigns/instructions/CoopBot.pdf>
2. **Droid Bot M (Modular)** — Classic driving base
<https://primelessons.org/RobotDesigns/instructions/DroidBotMSpikePrime.pdf>
3. **Droid Bot E (Expansion)** — Uses expansion set
<https://primelessons.org/RobotDesigns/instructions/DroidBotEBuildInstructions.pdf>
4. **Droid Bot IV** — Popular competition base
<https://primelessons.org/RobotDesigns/instructions/DroidBotIVBuildInstructions.pdf>
5. **ADB (Modified)** — All-purpose driving base
<https://primelessons.org/RobotDesigns/instructions/ADBModifications.pdf>
6. **Tricky Grab and Lift** — Grabber arm build
<https://primelessons.org/RobotDesigns/instructions/TrickyGrabandLift.pdf>
7. **Tic-Tac-Toe Bot** — Plays tic-tac-toe
<https://primelessons.org/RobotDesigns/instructions/tic-tac-toe.pdf>
8. **Earthquake Simulator** — Shake-table machine
<https://primelessons.org/RobotDesigns/instructions/SEarthquakeSimulator.pdf>
9. **Print and Scan** — Printer/scanner build
<https://primelessons.org/RobotDesigns/instructions/PrintandScan.lms>

2. Advanced / show-off builds

Cooler, harder projects from Pybricks — good "wow" goals to work toward.

1. **Groceries (machine learning)** — Sorts objects with AI camera
<https://pybricks.com/project/machine-learning-groceries/>
2. **Robotic arm with vision** — Camera-tracking arm
<https://pybricks.com/project/robot-arm-camera-tracker/>
3. **Ball balancer** — Balances on top of a ball
<https://pybricks.com/project/lego-spike-ball-balancer/>
4. **Hub menu launcher** — Run many programs from a menu
<https://pybricks.com/project/spike-hub-menu/>

3. More builds — Prof Bricks (themed & advanced)

Themed builds (medical, animals, arms) from Prof Bricks / Benedettelli.

1. **Powered Exoskeleton** — Medical theme
<https://robotics.benedettelli.com/spike-prime-powered-exoskeleton/>
2. **Balancing Wheelchair** — Medical theme
<https://robotics.benedettelli.com/spike-prime-balancing-robot/>
3. **Scanner Hat** — Medical theme
<https://robotics.benedettelli.com/spike-prime-scanner-hat/>
4. **Robot Arm + Joystick** — Surgical / pick-and-place
<https://robotics.benedettelli.com/spike-prime-pick-and-place-robot-joystick/>
5. **Adaptive Claw** — Grabber with jaw
<https://robotics.benedettelli.com/spike-prime-adaptive-claw/>
6. **T-Shirt Folding Robot** — Folds a shirt
<https://robotics.benedettelli.com/spike-prime-t-shirt-folding-robot/>
7. **SPIKE Prime Cat** — Animal build
<https://robotics.benedettelli.com/spike-prime-cat/>
8. **Brick-Sorting Bird** — Animal + sorting
<https://robotics.benedettelli.com/spike-learning-brick-sorter-bird-2/>
9. **Walking Bug (Hexapod)** — Six-legged walker
<https://robotics.benedettelli.com/spike-prime-bug-hexapod/>
10. **Guide Robot** — Rolling helper bot
<https://robotics.benedettelli.com/spike-prime-guide-robot/>
11. **Self-Driving Car** — Autonomous car
<https://robotics.benedettelli.com/spike-prime-car/>
12. **Color-Sorting Robot** — Sorts by color
<https://robotics.benedettelli.com/spike-prime-color-sorter/>

Sources: primelessons.org · pybricks.com/project/spike-prime · robotics.benedettelli.com (SPIKE Prime).
Printable companion to the interactive gallery at borenw.github.io/spike_prime_projects/